

Aws – Jovac 2026

LECTURE 1

- Introduction to Cloud
- Creating Aws account
- Introduction to Availability Zones and Regions in Aws
- To check availability zones in regions (Go to VPC and then Subnets)

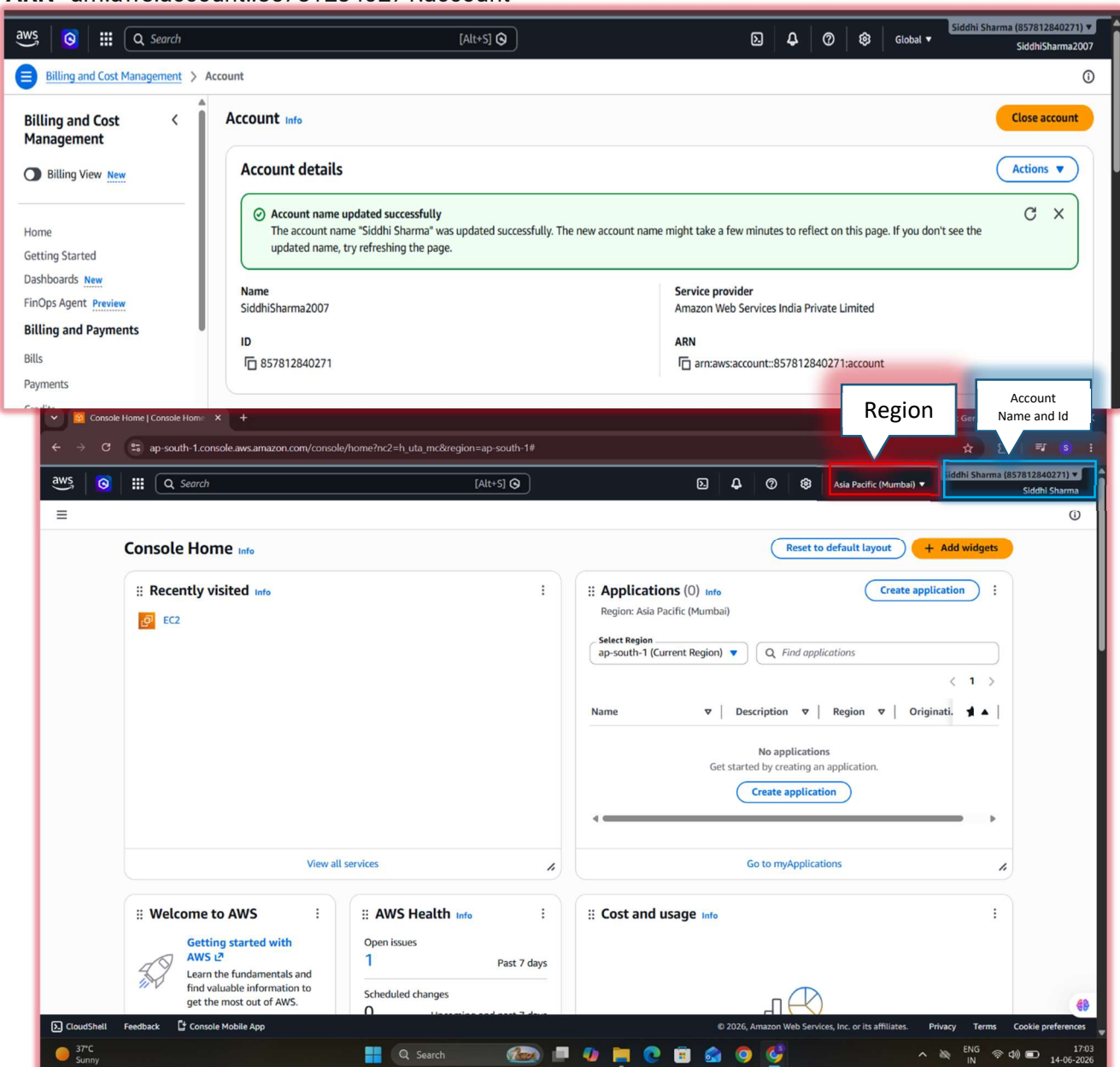
ACCOUNT DETAILS

Name- Siddhi Sharma

ID- 857812840271

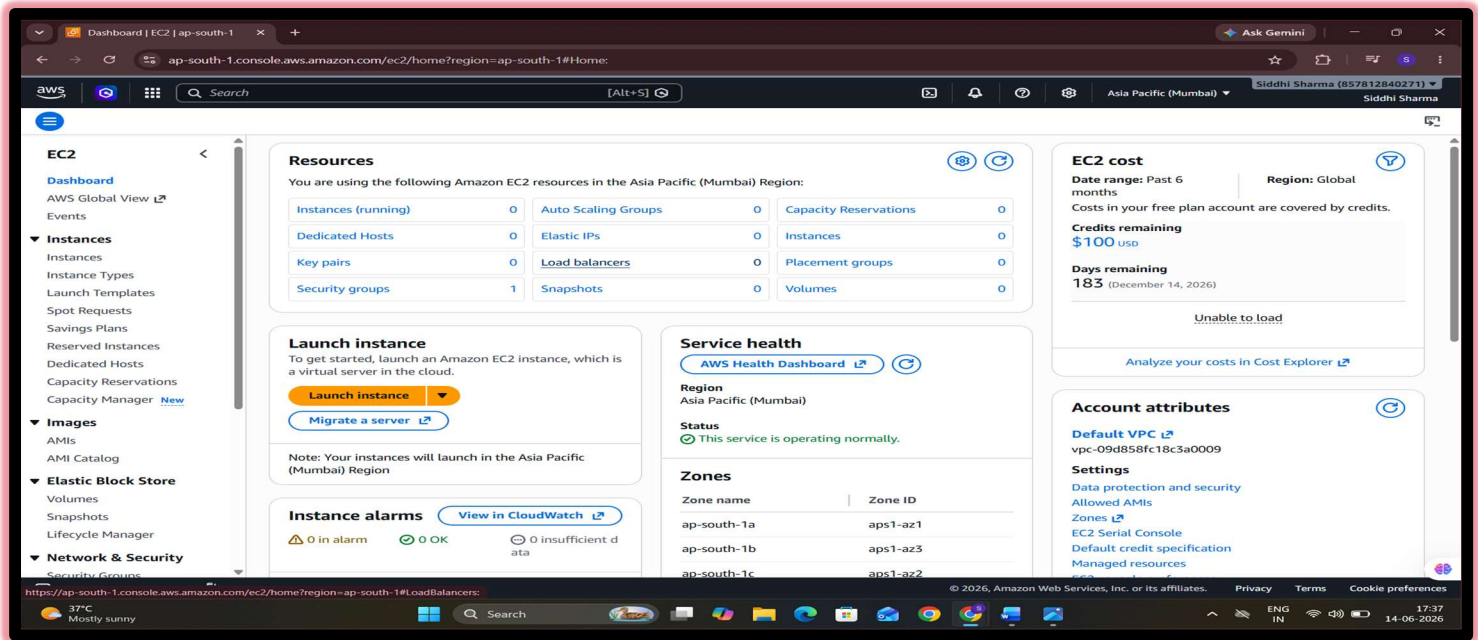
Service provider- Amazon Web Services India Private Limited

ARN- arn:aws:account::857812840271:account

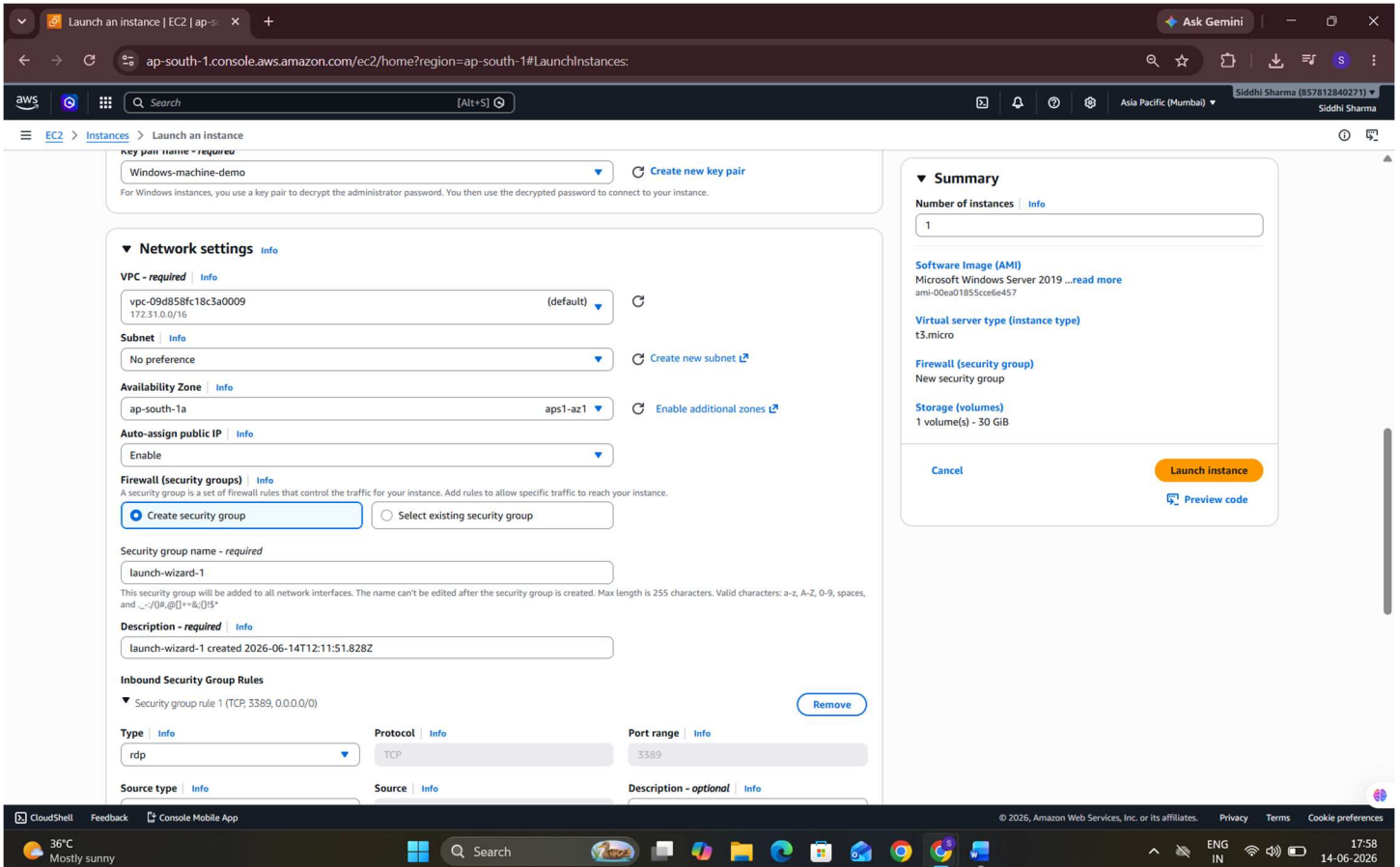


Task 1 – Create and Access Windows Machine on EC2 (Elastic Compute Cloud)

1. Select proper Region according to requirements.
2. Search for EC2 Service.
3. EC2 dashboard will open.



4. Click Launch Instance . Define Server name -> Select os (Windows) -> Select Version(Microsoft Windows Server 2019 Base) -> Select Instance Type(Ram and Processor)(t3.micro)



- Key Pair (Region specific service) – Used for connecting securely (authenticate) On creating key pair **Private key is downloaded to laptop** and **public key is stored in aws** and while connecting private key is used to authenticate during connection.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
Windows-machine-demo
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair (Not supported for Windows instances)

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

Windows-machine-demo.pem
1,678 B • Done

- Network Settings Select availability Zone 1a
- Create new security group for rules setting -> allow public ip so anyone can access it -> RDP is for remote desktop connection for accessing the instance (you can add custom group) Here we add http rdp

Launch instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Key pair name - required
Windows-machine-demo
For Windows Instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Network settings

VPC - required
vpc-09d858fc18c3a0009 (default)

Subnet
No preference

Availability Zone
ap-south-1a

Auto-assign public IP
Enable

Firewall (security groups)
Create security group

Security group name - required
launch-wizard-1

Description - required
launch-wizard-1 created 2026-06-14T12:11:51.828Z

Inbound Security Group Rules
Security group rule 1 (TCP, 3389, 0.0.0.0/0)

Summary
Number of instances: 1
Software Image (AMI): Microsoft Windows Server 2019
Virtual server type (instance type): t3.micro
Firewall (security group): New security group
Storage (volumes): 1 volume(s) - 30 GiB

[Launch instance](#)

8. Configure Storage – 30 gb SSD
9. Select Number of instances to be launched
10. Click Launch Instance (instance Started Showing Running)

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The 'Instances' page is active, displaying a list of EC2 instances. A filter is applied: 'Instance state = running'. One instance is listed: 'windows-machine' with ID 'i-01bcdfa585b63920', which is in the 'Running' state. The console includes a sidebar with navigation links like 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', 'AMIs', and 'AMI Catalog'. The top navigation bar shows the user 'Siddhi Sharma' and the region 'Asia Pacific (Mumbai)'.

11. Can be accessed using public ipv4 dns or public ip

This screenshot provides a detailed view of the EC2 instance 'i-01bcdfa585b63920' (Windows-Machine-Demo). The 'Details' tab is selected, showing various configuration parameters. Key details include:

- Instance summary:** Instance ID 'i-01bcdfa585b63920', IP v6 address, Hostname type 'IP name: ip-172-31-47-94.ap-south-1.compute.internal', Answer private resource DNS name 'IP v4 (A)', Auto-assigned IP address '13.233.233.29 [Public IP]', IAM role, IMDSv2 'Required', and Operator.
- Networking:** Public IPv4 address '13.233.233.29', Instance state 'Running', Private IP DNS name 'ip-172-31-47-94.ap-south-1.compute.internal', Instance type 't3.micro', VPC ID 'vpc-09d858fc18c3a0009', Subnet ID 'subnet-017b373c6dc401498', and Instance ARN 'arn:aws:ec2:ap-south-1:857812840271:instance/i-01bcdfa585b63920'.
- Public DNS:** 'ec2-13-233-233-29.ap-south-1.compute.amazonaws.com'.
- Private IPv4 addresses:** '172.31.47.94'.
- Elastic IP addresses:** None listed.
- AWS Compute Optimizer finding:** 'Opt-in to AWS Compute Optimizer for recommendations. | Learn more'.
- Auto Scaling Group name:** None listed.
- Managed:** 'false'.

12. You can modify security groups after creating and launching instance (added https)

The screenshot shows the AWS Management Console interface. A green notification banner at the top states: "Inbound security group rules successfully modified on security group (sg-0556a8e421736a9e1 | Windows-Machine-SG)". Below this, the "Details" section for the security group is displayed. The "Inbound rules" tab is selected, showing a table of three rules: HTTP on port 80, RDP on port 3389, and HTTPS on port 443. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security.

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images**
 - AMIs
 - AMI Catalog
- Elastic Block Store**
 - Volumes
 - Snapshots
 - Lifecycle Manager
- Network & Security**
 - Security Groups
 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces

sg-0556a8e421736a9e1 - Windows-Machine-SG

Details

Security group name Windows-Machine-SG	Security group ID sg-0556a8e421736a9e1	Description This is demo SG for windows machine	VPC ID vpc-09d858fc18c3a0009
Owner 857812840271	Inbound rules count 3 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (3)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0e1f5fc6ccb45ed40	IPv4	HTTP	TCP	80
-	sgr-067b15c9350f190ef	IPv4	RDP	TCP	3389
-	sgr-02d85c085b6b50dd3	IPv4	HTTPS	TCP	443

13. Add tags

The screenshot shows the AWS Management Console interface. A green notification banner at the top states: "Security Group tags updated. Security Group sg-0556a8e421736a9e1". Below this, the "Details" section for the security group is displayed. The "Tags" tab is selected, showing a table with two tags: "Name" with value "Windows-Machine-Demo" and "Purpose" with value "Demo". The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security.

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images**
 - AMIs
 - AMI Catalog
- Elastic Block Store**
 - Volumes
 - Snapshots
 - Lifecycle Manager
- Network & Security**
 - Security Groups
 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces

sg-0556a8e421736a9e1 - Windows-Machine-SG

Details

Security group name Windows-Machine-SG	Security group ID sg-0556a8e421736a9e1	Description This is demo SG for windows machine	VPC ID vpc-09d858fc18c3a0009
Owner 857812840271	Inbound rules count 3 Permission entries	Outbound rules count 1 Permission entry	

Tags

Key	Value
Name	Windows-Machine-Demo
Purpose	Demo

Accessing the Instance

1. We need password to access the machine.
2. We need key pair file (private key downloaded)
3. Select instance and then click connect

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The 'Instances' page is active, displaying a list of EC2 instances. The instance 'Windows-Machine-Demo' (ID: i-01bcfdfa585b63920) is selected. The 'Connect' button is visible. The instance details show it is in the 'Running' state with a public IPv4 address of 13.233.233.29.

4. Go to RDP client -> scroll down click on get password(username is shown there)

The screenshot shows the 'Connect to instance' page in the AWS Management Console. The 'RDP client' tab is selected. The 'Get password' button is visible. The page shows the instance details and the connection type options.

5. upload private key downloaded earlier -> click decrypt password

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-01bcdfa585b63920 (Windows-Machine-Demo)

Key pair associated with this instance
Windows-machine-demo

Private key
Either upload your private key file or copy and paste its contents into the field below.

Upload private key file

Windows-machine-demo....
1.68 KB

Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEA/vYJCTypPH4G2FDDqHdIsiCrvRhvTgb9rq71r5Wilmijz6
KZPKqIIQO6hECKlqULk07wBEozDUT4ciGE+FIHhwAeKTVkqebqM5+71RP6QT48O
2a2gfNDErknkBKh2Tv52kUbBeZenyYuyKEVSWXgp1/tVclyKQRKvE25vg+vYNmFL
3kofa8wO+7lUgUQlhr8L3wF85CxL2i1tUoyZ6+uccGF6ivvv+uLBO/ewPC5onzDq
g/Et/6EftaDZ5x+1kWzIJOg/MoRG1zX/PvP/8e9JA5GALAmneVE55p2hwkX0sBgG
vtOKTnIodZhVkcq+u7g7YSL8hf/JvCvg34zHGQIDAQABAoIBADx//GT9iVW/ZMbZ
IEf8ZjmlGhpy42pXiUsGBnj22pZmdxYEOUt6+2j5Gf3+cv7AySiAMUfah3Rw1Aj
-----
```

Cancel Decrypt password

6. Password is now visible

SSM Session Manager RDP client EC2 serial console

Instance ID
i-01bcdfa585b63920 (Windows-Machine-Demo)

Connection Type

Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS
ec2-13-233-233-29.ap-south-1.compute.amazonaws.com

Username
Administrator

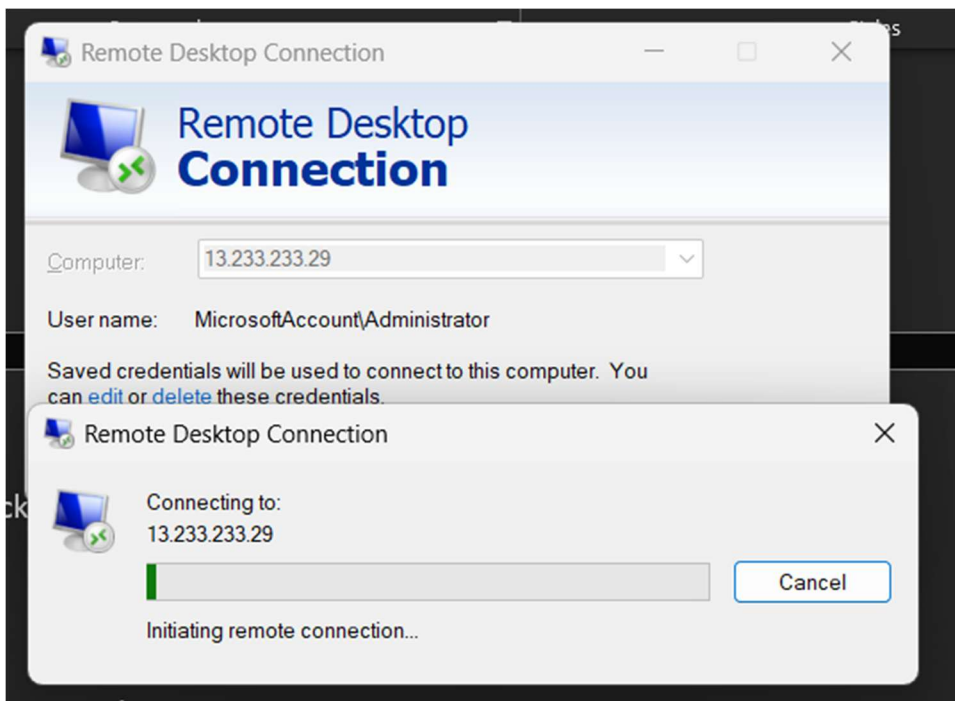
Password
x=PnkVdsGK97yghGz5UV4n5b9yO-5V@T

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

7. Access using public DNS

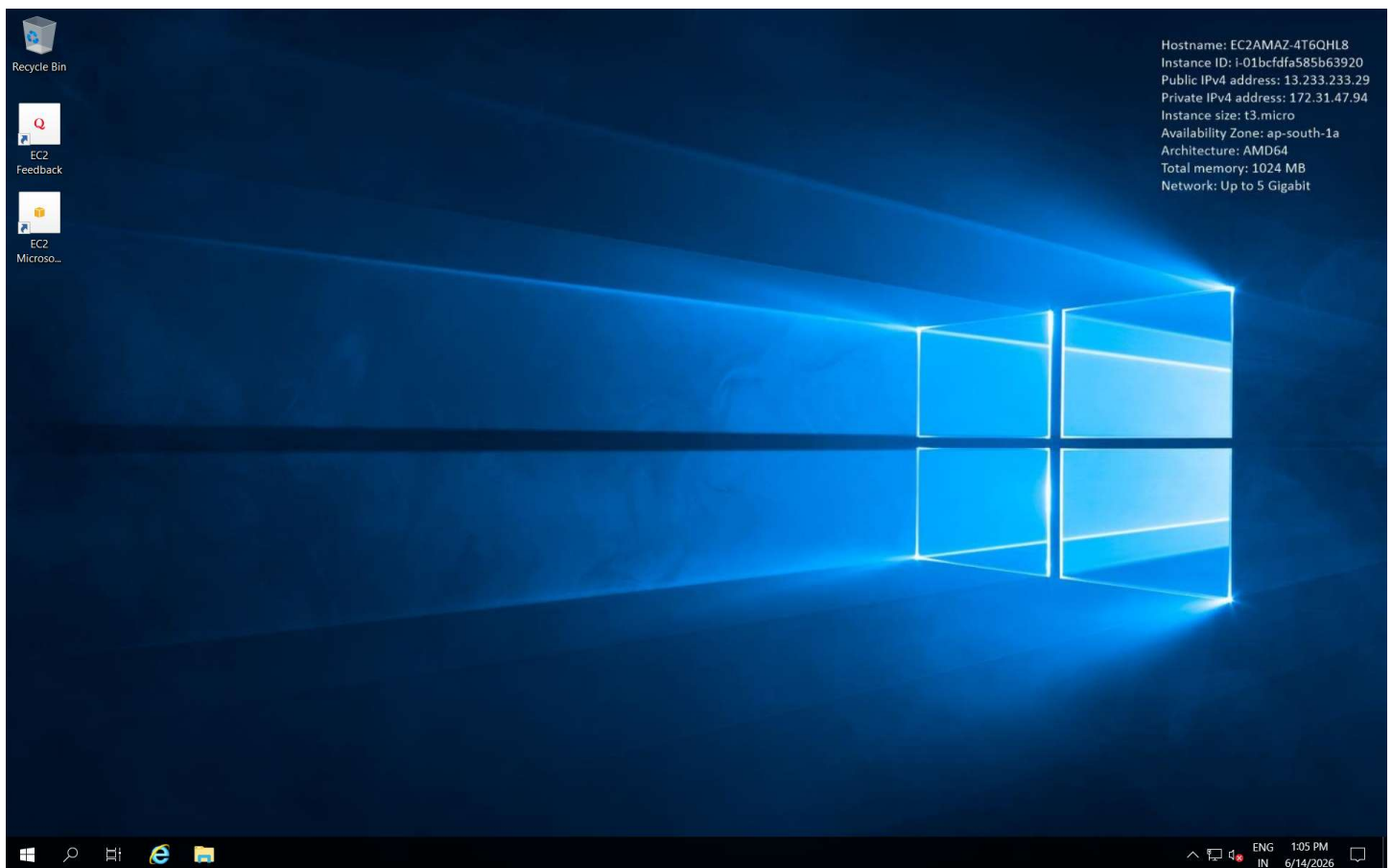
8. Win + R to open run box->type mstsc(Microsoft Terminal Service) in run box to open remote desktop connection

9. Enter public IP and click connect

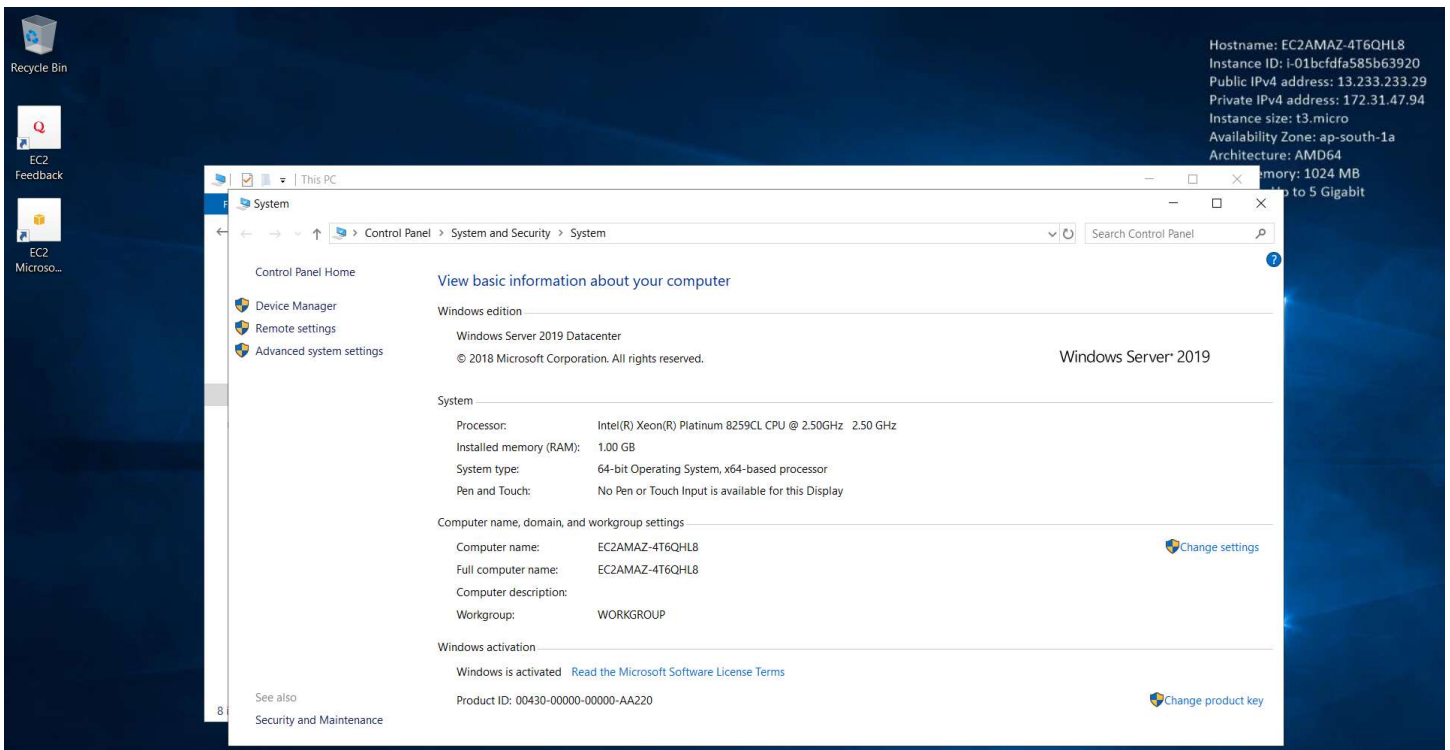


10. Enter username and password

11. Server is accessed



Properties of server



12. Terminate After Use

